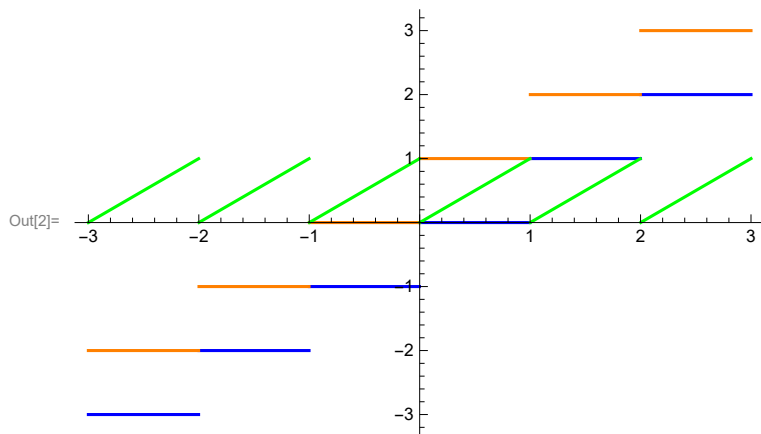
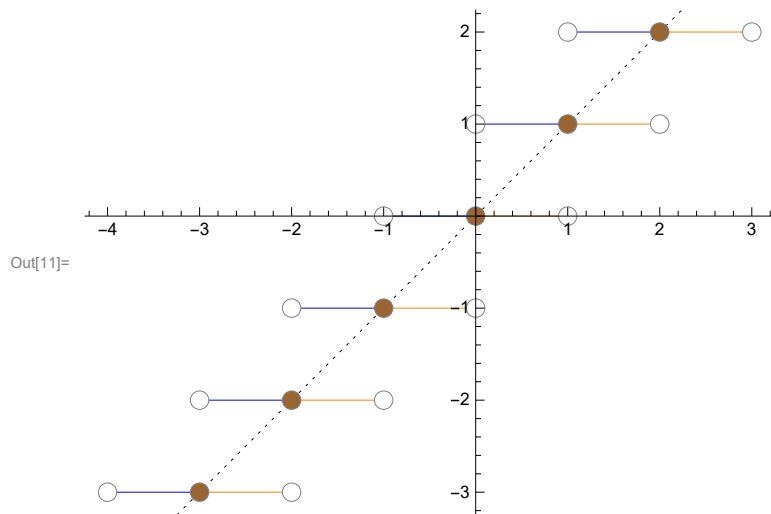


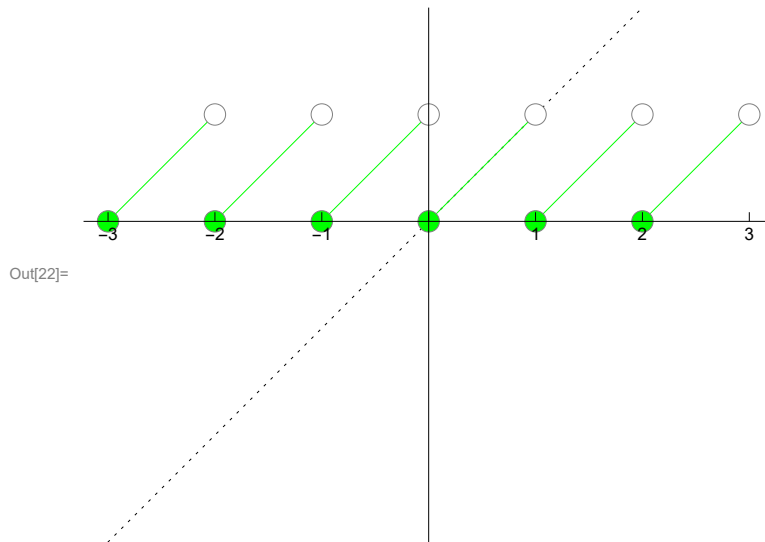
In[2]:= **Plot**[{**Floor**[x], **Ceiling**[x], **Mod**[x, 1]}, {x, -3, 3}, **PlotStyle** → {Blue, Orange, Green}]



In[11]:= **Graphics**[{{**Dotted**, **InfiniteLine**[{0, 0}, {1, 1}]},
EdgeForm[Gray], {Blue, **Table**[**Line**[{{x - 1, x}, {x, x}}], {x, -3, 2}]},
{Orange, **Table**[**Line**[{{x, x}, {x + 1, x}}], {x, -3, 2}]},
{Brown, **Table**[**Disk**[{x, x}, .1], {x, -3, 2}]},
{White, **Table**[{**Disk**[{x - 1, x}, .1], **Disk**[{x + 1, x}, .1]}, {x, -3, 2]}}], **Axes** → **True**]



```
In[22]:= Graphics[{{Dotted, InfiniteLine[{0, 0}, {1, 1}]},
  EdgeForm[Gray], {Green, Table[Line[{{x, 0}, {x + 1, 1}}], {x, -3, 2}],
  Table[Disk[{x, 0}, .1], {x, -3, 2}], {White, Table[{Disk[{x + 1, 1}, .1]}, {x, -3, 2}]}],
  Axes → True, Ticks → {Range[-3, 3], {}}, PlotRange → {-3, 2}]
```



Floor and Ceiling

Example:

$$\lfloor 10.3 \rfloor = 10$$

$$\lfloor -10.3 \rfloor = -11$$

Identities:

$$\lfloor x \rfloor = x \text{ iff } x \text{ is an integer}$$

$$\lfloor x \rfloor \leq x$$

$$x - 1 < \lfloor x \rfloor$$

$$\lceil x \rceil - \lfloor x \rfloor = 0 \text{ iff } x \text{ is an integer, otherwise } 1$$

$$-\lceil x \rceil = \lfloor -x \rfloor$$

$$\lfloor x + n \rfloor = \lfloor x \rfloor + n \text{ iff } n \text{ is an integer}$$

Usually by the integer part, it is meant $\lfloor x \rfloor$

The difference: $x - \lfloor x \rfloor$ is then called the fractional part.

Note: other conventions may exist for negative numbers.

True or False: $\lfloor x + y \rfloor = \lfloor x \rfloor + \lfloor y \rfloor$

What is the difference between

$$\lfloor x \rfloor + 1$$

and

$$\lceil x \rceil$$

Which one is used to find the number of bits needed to express n in binary?

$$\lfloor \log_2 n \rfloor$$

$$\log_2 \lfloor n \rfloor$$

$$\lfloor \log_2 n \rfloor + 1$$

$$\lceil \log_2 n \rceil$$

Simplify

$$\lfloor \sqrt{\lfloor x \rfloor} \rfloor$$

$$\lceil \lfloor x \rfloor \rceil$$

How many integers are in the interval $[a,b]$?

Mod

This is similar to the fractional part except one gives two arguments.

$x \bmod b$

b times the fractional part of x/b

Since the fractional part is in $[0,1)$ it follows that

$$y = x \bmod b$$

satisfies $0 \leq y < b$

Usually mod is just for integer arguments.

Mod is useful for periodic functions.

Out[25]=

n	f(n)
0	0
1	1
2	2
3	0
4	1
5	2
6	0
7	1
8	2
9	0
10	1

Mod also gives the ones-digit in base b :

Out[27]=

n	n mod 10
18	8
19	9
20	0
21	1
22	2
23	3
24	4
25	5